

HAMZA EL HADDAD SABRI

Cybersecurity Engineering Student | Pentesting · Network Security · Malware Analysis · Defensive Security · HPC

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PROFESSIONAL SUMMARY

Cybersecurity Engineering student at ENTI–Universitat de Barcelona with hands-on experience across the full offensive and defensive security lifecycle: passive and active reconnaissance, OSINT gathering, port scanning and service enumeration, CVE-based vulnerability research, exploitation with Metasploit and manual techniques, and network traffic analysis and intrusion detection with Suricata IDS/IPS. Builder of real security tools including a unified pentesting platform (ShadowScan), a graph-based lateral movement simulator (RedTrace), and GPU-accelerated cryptanalysis implementations using CUDA and OpenMP. Proficient with Wireshark, Nmap, Kali Linux, Suricata, and Metasploit, with active self-directed practice on TryHackMe. Highly proficient at leveraging AI tools to accelerate security research, threat intelligence gathering, and code review workflows. Collaborative, fast-learning, and deeply motivated to grow within a professional cybersecurity team.

TECHNICAL SKILLS

Programming Languages

C / C++, Python, Java, Bash, Go, SQL

Cryptography

RSA, AES, Hash Functions, Digital Signatures, PKI, Symmetric & Asymmetric encryption, Key management

Offensive Security & Pentesting

Reconnaissance (passive/active), OSINT, Nmap, Suricata IDS/IPS (rule writing & tuning), Snort, SIEM Nessus/OpenVAS, CVE research & exploitation, concepts, log analysis, network monitoring, incident Metasploit, privilege escalation, lateral movement response, server hardening simulation

Defensive Security

Network Security & Attacks

DoS/DDoS, SYN Flood, HTTP Flood, ARP Spoofing, Wireshark, Tcpdump, Scapy, Metasploit, Burp Suite, OSI-layer attack analysis, Firewall configuration, Nmap, Nessus, Kali Linux, TryHackMe
VLANs, NAT, DHCP, DNS, Routing & Switching

Security Tools

HPC & Parallel Computing

OpenMP (shared memory), MPI (distributed clusters), AWS / Azure fundamentals, IAM policies, security groups, cloud hardening, S3 security, cloud-native security principles
CUDA / GPGPU programming, brute-force cryptanalysis

Cloud & DevSecOps

OS & Scripting

Linux (Arch, Debian, Kali), systemd, Bash & Python automation, log parsing, process management, forensic investigation basics
MySQL, PostgreSQL, SQLite · Advanced SQL: Triggers, Views, Stored Procedures, normalization (3NF), SQL injection awareness

Databases

GRC & Compliance

ISO 27001, ENS, GDPR, NIST · Risk management, threat modeling, compliance auditing, security policy design
Advanced use of AI tools (ChatGPT, Claude, Copilot) for security research, code review, threat intelligence, and report generation

AI & Productivity

Malware Analysis

Static & dynamic analysis, obfuscation techniques, behavioral analysis, malware typology, sandbox usage
Strong team player · Fast self-directed learner · Analytical problem solver · Proactive communicator · Adaptable under pressure

Soft Skills

PERSONAL PROJECTS & PORTFOLIO

RedTrace — Lateral Movement Network Simulator

2024–2025

Python · Graph Theory · Dijkstra · github.com/HamzaEnti/RedTrace

- Simulates attacker lateral movement across enterprise networks using weighted directed graphs and Dijkstra's shortest-path algorithm to identify optimal attack routes between hosts
- Models real-world TTPs (MITRE ATT&CK) including privilege escalation, network segmentation bypass, and multi-hop pivoting through internal subnets
- Enables red team exercises and proactive internal security posture evaluation by mapping all reachable attack paths from a given entry point

ShadowScan — Unified Penetration Testing Platform

2024–2025

Python · GUI · Security Tooling · github.com/HamzaEnti/ShadowScan

- Full-featured pentesting platform with graphical interface consolidating port scanning, service fingerprinting, and vulnerability discovery into a single professional application
- Supports reconnaissance and enumeration workflows, integrating Nmap-compatible scanning with structured, exportable result reporting
- Reduces operational overhead for security auditors by centralizing tool output, enabling efficient and repeatable network audits

Parallel-Programming-C — HPC & Cryptanalysis

2024–2025

C · OpenMP · MPI · CUDA · github.com/HamzaEnti/Parallel-Programming-C

- Optimized C implementations of parallel algorithms for HPC using OpenMP (shared memory) and MPI (distributed clusters)
- GPU-accelerated brute-force password cracking with CUDA, demonstrating applied cryptanalysis and GPGPU performance optimization for security workloads

go-raft-consensus — Distributed Systems

2024–2025

Go · Consensus Algorithms · github.com/HamzaEnti/go-raft-consensus

- From-scratch implementation of the Raft consensus algorithm in Go, covering leader election, heartbeat mechanisms, and fault-tolerant log replication
- Demonstrates deep understanding of distributed system design and consensus protocol engineering relevant to cloud-native security architectures

RainbowWarriors — Secure Password Manager

2024–2025

Python · AES Cryptography · GUI · github.com/HamzaEnti/RainbowWarriors

- Password manager implementing AES encryption/decryption with secure key derivation and intuitive GUI, applying applied cryptography best practices throughout
- Designed with defense-in-depth principles: master key protection, data integrity validation, and secure in-memory credential handling

ZalandoBot — Browser Automation & Web Scraping

2024

JavaScript · Chrome Extension API · github.com/HamzaEnti/ZalandoBot

- Chrome extension demonstrating DOM manipulation, real-time product data extraction, and browser automation — skills directly applicable to web application security research and XSS/CSRF vulnerability analysis

Cybersecurity Year Recap 2024–2025 — Academic Portfolio

2025

Python · C · SQL · Networking Tools · github.com/HamzaEnti/Cybersecurity-Year-Recap-2024-2025

- Comprehensive portfolio covering malware analysis labs, network design, database security, and system administration developed across the 2024–2025 academic year

EDUCATION

Grado en Ingeniería de la Ciberseguridad

Sep 2024 – Present

ENTI — Universitat de Barcelona

- Outstanding academic record — Risk Analysis & Management (10.0), Networks III (9.61), Governance & Standards (8.91), Networks II (8.25), Advanced Algorithms (8.09), Cryptography (7.60), Malware Analysis (7.68)
- Core areas: Advanced Cryptography, Network Attack Techniques, Malware Typology & Analysis, AI Applied to Cybersecurity, Parallel & Distributed Systems, Risk Management, GRC & Compliance, Access Control, Database Security

Grado en Ingeniería Telemática

Sep 2023 – Jul 2024

Universitat Pompeu Fabra

- Foundation in telecommunications: TCP/IP protocol stack, OSI model, signal theory, and network architecture design

Bachillerato Tecnológico

Sep 2021 – May 2023

La Salle Manresa

PROFESSIONAL EXPERIENCE

Computer Science Instructor

Sep 2021 – Sep 2022

Pensactius · Manresa

- Delivered theoretical and practical training in computer fundamentals, hardware maintenance, LAN configuration, and basic security practices to groups of students
- Developed custom instructional materials; provided Level 1 technical support resolving software and hardware incidents under time pressure
- Managed classroom IT infrastructure: user account administration, network setup, and system maintenance — developing systematic problem-solving and communication skills

LANGUAGES & SELF-LEARNING

Languages: Spanish (Native) · Catalan (Native) · Arabic (Native) · English (Advanced — C1)

Platforms & Certifications: TryHackMe (Active — cybersecurity challenges & CTF) · Python for Data Science, Udemy (2022) · HTML5 & CSS, W3Schools (2022)